

Vishvam Hitendrakumar Patel

AI/ML Engineer/Data Scientist

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Professional Summary

AI/ML Engineer/Data Scientist with around 2 years of experience in designing, deploying, and scaling end-to-end machine learning solutions across NLP, computer vision, generative AI (LLMs), and advanced analytics, leveraging deep learning and statistical modeling. Proficient in building supervised and unsupervised models using algorithms like Logistic Regression, Random Forest, XGBoost, SVM, KMeans, and DBSCAN, optimized for high performance and scalability. Skilled in fine-tuning and deploying architectures such as CNNs, LSTMs, Transformers, and large language models like BERT, GPT, and LLaMA using frameworks including TensorFlow, PyTorch, Hugging Face, and Keras. Strong expertise in Python programming, data engineering, feature engineering, and data visualization using Pandas, NumPy, Scikit-learn, OpenCV, NLTK, SpaCy, Matplotlib, and Seaborn. Experienced in ML infrastructure and MLOps practices, deploying production-grade AI systems on AWS using tools like Docker, Kubernetes, Terraform, and CI/CD pipelines.

Education

Master of Science in Computer Science | University of Alabama in Huntsville, Alabama, USA

Skills

Languages / IDEs: Python, Jupyter Notebook, Google Colab, VS Code, SSMS

Machine Learning: Linear Regression, Logistic Regression, Decision Trees, Random Forests, Naive Bayes, A/B Testing, NumPy

Deep Learning: CNN, LSTM, Natural Language Processing (NLP), Large Language Models (LLM), Hugging Face Transformers (BERT, GPT-3), LangChain

Cloud / Visualization: AWS (EC2, SQS, SNS, CodeDeploy, CloudWatch, API Gateway), Tableau, Power BI

Statistical Techniques: Hypothesis Testing, Data Visualization, Data Modelling, Model Evaluation

Packages & Frameworks: Pandas, NumPy, Matplotlib, Scikit-learn, Seaborn, TensorFlow, Keras, XGBoost, PyTorch, NLTK

Databases & Tools: SQL Server, MySQL, PostgreSQL, Redis

Work Experience

American Red Cross, Washington DC

May 2024-Present

AI/ML Engineer

- Designed and fine-tuned BERT-based NLP models in PyTorch to process over 1.2 million legal documents annually, enabling faster case preparation and improving efficiency for compliance teams across multiple US states.
- Developed TensorFlow-powered CNN pipelines for medical imaging that reduced diagnosis review time from 4 minutes to under 20 seconds for over 500,000 patient scans per year, supporting faster clinical decision-making.
- Built large-scale recommendation systems using ensemble models in Scikit-learn that generated over 4.5 million personalized recommendations annually, enhancing user engagement and supporting data-driven decision-making across operations.
- Implemented end-to-end MLOps pipelines on AWS SageMaker, Docker, and Kubernetes, enabling over 120 production model deployments with zero downtime, improving release confidence and scalability for enterprise AI systems.
- Crafted anomaly detection frameworks using Autoencoders to identify over 8,200+ irregular patient claims annually, safeguarding healthcare providers from fraudulent billing activities and reducing operational losses.
- Migrated legacy ML models to AWS with TensorFlow Serving, increasing processing throughput from 1,200 to over 3,000+ API requests per second, ensuring consistent and reliable performance during sustained peak load conditions.
- Automated ETL workflows with Pandas, NumPy, and Apache Spark, reducing data processing runtime from 5 hours to just 50 minutes across pipelines handling 2 TB of daily structured and unstructured datasets.
- Coordinated with cross-functional teams to deploy Hugging Face Transformer models that processed over 25 million customer interactions annually, enabling targeted marketing campaigns and improving customer engagement quality.

Alabama Baptist State Board of Missions, Prattville, Alabama

Apr 2023– Aug 2023

Data Scientist

- Fabricated and deployed predictive analytics models to forecast event attendance, improving planning accuracy for over 300 statewide outreach programs annually, helping resource allocation and volunteer scheduling become more efficient.
- Architected natural language processing pipelines to analyze sermon transcripts and training materials, enabling content categorization across 25,000+ archived resources for better digital accessibility and faster educational content delivery to local churches.
- Constructed image classification models to tag and organize mission trip photos, processing over 120,000 media files, which significantly enhanced the organization's historical records management and publication workflows.
- Created data-driven donation trend dashboards in Power BI, enabling leadership to identify seasonal giving patterns, uncover regional participation gaps, and support targeted donor engagement campaigns for sustainable funding growth.
- Streamlined reporting workflows with Python, Pandas, and Advanced Excel, reducing manual reporting time from 10 hours to under 4 hour per week, improving accuracy and allowing focus on strategic decision-making.
- Collaborated with the communications team to implement text sentiment analysis on community feedback surveys, processing 15,000+ responses annually to highlight improvement areas and strengthen outreach program effectiveness.
- Integrated deep learning models using TensorFlow to enhance video captioning accuracy for online sermons, ensuring accessibility compliance, increasing viewer retention, and supporting engagement among hearing-impaired audiences.

Projects

Multimodal Sentiment Analysis (Skills Used: *Hugging Face Transformers, PyTorch, Torchvision, Torchaudio, OpenCV, Pillow, librosa, NLTK, SpaCy*)

- Established a multimodal sentiment analysis system integrating image, audio, and text data using Hugging Face Transformers for NLP, CNNs with Torchvision for image processing, and Torchaudio with librosa for audio feature extraction, improving sentiment detection accuracy.
- Devised and deployed an end-to-end inference pipeline leveraging PyTorch for training, FastAPI for real-time predictions, and modality-specific preprocessing using OpenCV, Pillow for images, and NLTK, SpaCy for text, ensuring seamless feature fusion via a web interface.

Lumbar Spine Degenerative Classification (Skills Used: *PyTorch, MONAI, OpenCV, SimpleITK, pydicom, NumPy, Pandas*)

- Formed a multi-task deep learning model using PyTorch and MONAI to classify severity of five lumbar spine degenerative conditions from MRI scans, improving accuracy by 14% over baseline through advanced ordinal regression and multi-head architecture.
- Formulated an ROI extraction pipeline with OpenCV and SimpleITK based on radiologist annotations, reducing background noise and enhancing macro F1-score by 11%, while handling missing labels effectively using masked loss and semi-supervised learning techniques.